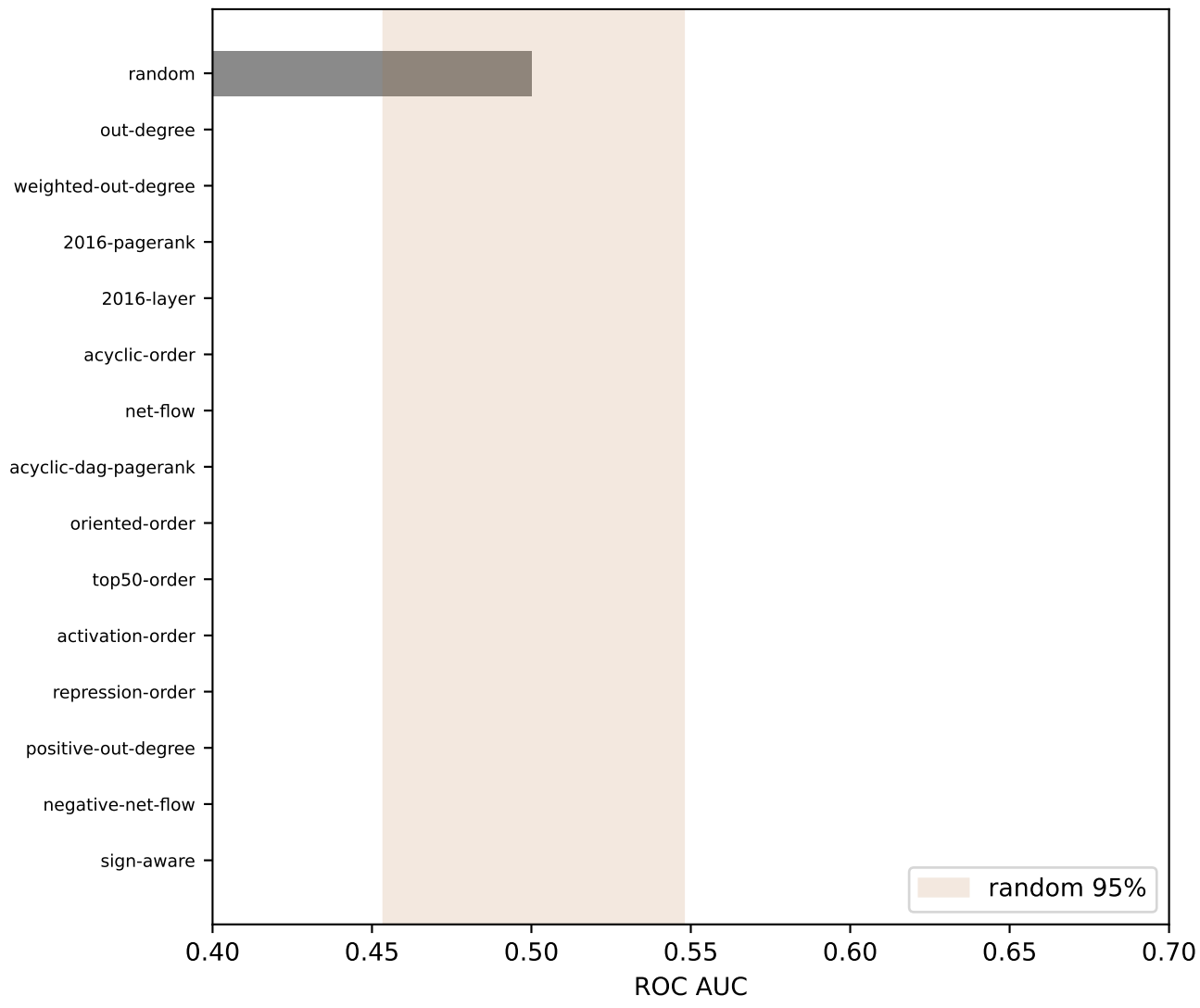
Regulator ranking against reference 'regulators'

2026-11-07 · 739 genes ranked, 199 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	26.8 (19–35)	0.500 (0.454–0.548)				
out-degree	0	0.215	1.000	1.000	-0.115 (-0.134, -0.098)	< 0.001
weighted-out-degree	1	0.330	1.000	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	10	0.294	1.000	1.000	-0.036 (-0.076, +0.005)	0.098
2016-layer	70	0.345	< 0.001	1.000	+0.016 (-0.022, +0.054)	0.444
acyclic-order	0	0.198	1.000	1.000	-0.132 (-0.163, -0.101)	< 0.001
net-flow	1	0.332	1.000	1.000	+0.002 (-0.009, +0.013)	0.692
acyclic-dag-pagerank	0	0.207	1.000	1.000	-0.123 (-0.146, -0.101)	< 0.001
oriented-order	2	0.166	1.000	1.000	-0.164 (-0.198, -0.129)	< 0.001
top50-order	17	0.269	0.995	1.000	-0.060 (-0.102, -0.015)	0.004
activation-order	3	0.240	1.000	1.000	-0.090 (-0.129, -0.052)	< 0.001
repression-order	3	0.238	1.000	1.000	-0.092 (-0.129, -0.057)	< 0.001
positive-out-degree †	1	0.274	1.000	1.000	-0.056 (-0.081, -0.034)	< 0.001
negative-net-flow †	2	0.278	1.000	1.000	-0.052 (-0.076, -0.028)	< 0.001
sign-aware †	1	0.266	1.000	1.000	-0.063 (-0.077, -0.051)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.

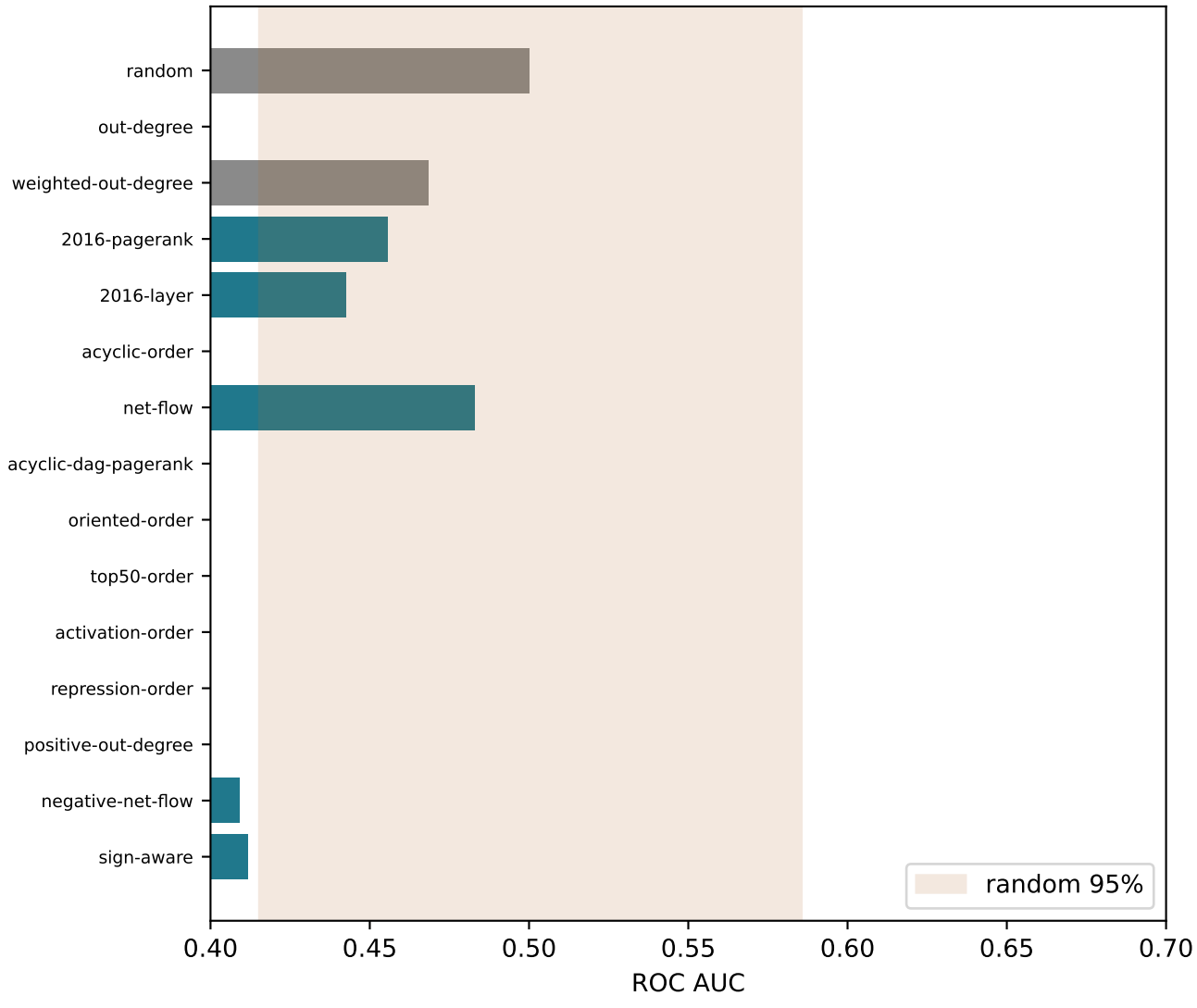


Regulator ranking against reference 'top-layer'

2026-11-07 · 739 genes ranked, 46 in top-regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	6.2 (2–11)	0.500 (0.415–0.586)				
out-degree	0	0.358	1.000	1.000	-0.110 (-0.144, -0.079)	< 0.001
weighted-out-degree	1	0.468	0.998	0.756	+0.000 (+0.000, +0.000)	
2016-pagerank	3	0.455	0.958	0.844	-0.013 (-0.093, +0.070)	0.712
2016-layer	21	0.442	< 0.001	0.905	-0.026 (-0.101, +0.046)	0.466
acyclic-order	0	0.334	1.000	1.000	-0.134 (-0.197, -0.078)	< 0.001
net-flow	1	0.483	0.998	0.661	+0.015 (-0.007, +0.037)	0.168
acyclic-dag-pagerank	0	0.341	1.000	1.000	-0.127 (-0.168, -0.089)	< 0.001
oriented-order	1	0.290	0.998	1.000	-0.179 (-0.261, -0.101)	< 0.001
top50-order	5	0.391	0.763	0.995	-0.077 (-0.158, +0.004)	0.058
activation-order	1	0.329	0.998	1.000	-0.139 (-0.207, -0.074)	< 0.001
repression-order	2	0.370	0.987	0.999	-0.098 (-0.167, -0.030)	0.004
positive-out-degree †	1	0.389	0.998	0.995	-0.080 (-0.123, -0.039)	< 0.001
negative-net-flow †	1	0.409	0.998	0.983	-0.059 (-0.105, -0.016)	0.006
sign-aware †	1	0.412	0.998	0.981	-0.057 (-0.080, -0.032)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

true

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	2,868	0.897 (60.35×)	69.20×	46.63×	1.00 (0.58)
resolve_reciprocals → drop_sinks → prune_triangles	2,137	0.116 (49.96×)	53.19×	46.17×	1.00 (0.51)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	972	0.111 (50.64×)	52.48×	48.65×	1.00 (0.50)
absolute	2,868	0.897 (60.35×)			
absolute → minimum_feedback_order	2,868	0.659 (58.13×)			
orient_reciprocals ratio=2.0 → absolute	2,868	0.569 (57.95×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	2,868	0.569 (57.95×)			
top_outgoing k=50 → absolute	2,868	0.897 (62.82×)			
top_outgoing k=50 → absolute → minimum_feedback_order	2,868	0.664 (59.93×)			
keep_sign sign=1 → absolute	2,868	0.626 (69.20×)			
keep_sign sign=1 → absolute → minimum_feedback_order	2,868	0.464 (66.63×)			
keep_sign sign=-1 → absolute	2,868	0.272 (46.63×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	2,842	0.199 (43.69×)			

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 38,659 edges (18,220 negative), largest strong component 739.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 38,659 edges (18,220 negative), largest strong component 739.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 7,432 edges (3,990 negative), largest strong component 167.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 7,432 edges (3,990 negative), largest strong component 167; DAG keeps 3,264 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 38,659 edges (0 negative), largest strong component 739; DAG keeps 33,676 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 38,659 edges (0 negative), largest strong component 739.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 38,659 edges (0 negative), largest strong component 739; DAG keeps 33,676 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 28,882 edges (0 negative), largest strong component 1; DAG keeps 28,882 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 27,567 edges (0 negative), largest strong component 739; DAG keeps 23,512 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 20,439 edges (0 negative), largest strong component 737; DAG keeps 17,730 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 18,220 edges (0 negative), largest strong component 730; DAG keeps 15,965 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 38,659 edges (18,220 negative), largest strong component 739.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 38,659 edges (18,220 negative), largest strong component 739.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 38,659 edges (18,220 negative), largest strong component 739.